



Olist Marketplace Performance Customer Experience Analysis

Data Analytics Hackathon

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Data Analytics Hackathon Submission

Project Overview

Olist Marketplace Performance & Customer Experience Analysis is a data analytics project focused on understanding the overall performance of the Olist e-commerce marketplace and identifying the key factors associated with customer satisfaction and dissatisfaction.

The analysis uses approximately **100,000 real e-commerce orders from Brazil**, covering the period from **September 2016 to October 2018**. The dataset includes information related to **orders, customers, sellers, products, payments, deliveries, reviews, and geographic locations**.

The project combines multiple business dimensions to create a consolidated view of marketplace performance. Key areas of analysis include **revenue and order trends, delivery performance, customer review scores, product category performance, geographic patterns, seller performance, payment behavior, and customer retention**.

A major focus of the project is identifying the factors most strongly associated with **low customer satisfaction**. Rather than assuming a single cause, the analysis evaluates multiple contributing factors—particularly **delivery delays, regional logistics challenges, seller performance, and product category differences**.

The ultimate goal is to transform raw marketplace data into **evidence-based business insights and actionable recommendations** that can help Olist improve the customer experience, optimize operational performance, and support sustainable marketplace growth.

Business Problem

Olist operates a large and growing e-commerce marketplace connecting customers, sellers, products, payment systems, and logistics partners across Brazil. As the marketplace expands, maintaining a consistent and high-quality customer experience becomes increasingly challenging.

The available data shows significant variation in **delivery performance, customer review scores, seller performance, product categories, payment behavior, and geographic regions**. However, these factors are interconnected, and Olist needs a clear understanding of **which factors are most strongly associated with customer satisfaction and dissatisfaction**.

The key business challenge is to move beyond isolated metrics and develop a consolidated, evidence-based understanding of marketplace performance.

The analysis aims to answer:

1. How is the marketplace performing in terms of **orders, revenue, and customer satisfaction**?
2. How strongly are **delivery delays associated with low customer review scores**?
3. Which **states and geographic regions** experience weaker operational performance?
4. Which **product categories** are performing strongly, and which require attention?
5. How does **seller performance** vary across the marketplace?
6. How do **payment methods and installment behavior** relate to customer purchasing patterns?
7. What does the data reveal about **customer retention and repeat purchasing**?
8. Which factors appear to be the **primary drivers of customer dissatisfaction**, and which are secondary contributors?.

Objective

The primary objective of this project is to develop a clear, evidence-based understanding of **Olist's marketplace performance and customer experience** using historical e-commerce data.

The analysis aims to identify the factors most strongly associated with **customer satisfaction and dissatisfaction** while evaluating performance across key business dimensions such as **orders, revenue, delivery, sellers, product categories, geography, payments, and customer behavior**.

Key Objectives

- **Evaluate Marketplace Performance**
Analyze order volume, revenue trends, and overall marketplace performance over time.
- **Assess Delivery Performance**
Examine how delivery timing and delays are associated with customer review scores and satisfaction.
- **Identify Customer Satisfaction Drivers**
Determine which factors appear to have the strongest relationship with low customer review scores.
- **Analyze Product Category Performance**
Identify high-performing, underperforming, and growth-opportunity product categories.
- **Evaluate Geographic Performance**
Compare marketplace performance, delivery experience, and customer satisfaction across Brazilian states.
- **Analyze Seller Performance**
Understand seller contribution, revenue concentration, and performance differences across the marketplace.
- **Understand Payment Behavior**
Analyze payment methods, installment usage, and their relationship with purchasing patterns.
- **Explore Customer Value and Retention**
Identify customer segments and evaluate repeat purchasing behavior.
- **Provide Actionable Recommendations**
Convert analytical findings into practical recommendations that can help Olist improve operations, customer experience, and sustainable marketplace growth.

Dataset Description

This project uses the **Brazilian E-Commerce Public Dataset by Olist**, containing approximately **100,000 orders** placed in Brazil between **September 2016 and October 2018**.

The dataset includes interconnected information on **customers, orders, products, sellers, payments, deliveries, reviews, and geographic locations**.

The data was cleaned, standardized, and merged to support a comprehensive analysis of **marketplace performance and customer experience**.

Data Preparation

Before analysis, the dataset was processed to ensure consistency and reliability. The preparation process included:

- Handling missing values
- Cleaning and standardizing data
- Removing or flagging invalid records
- Converting date columns into appropriate formats
- Creating derived business metrics
- Merging relevant datasets for cross-dimensional analysis

Dataset Purpose in This Project

The dataset provides the foundation for analyzing how different aspects of marketplace operations are associated with business performance and customer satisfaction. By connecting **delivery performance, review scores, product categories, sellers, geographic regions, payment behavior, and customer purchasing patterns**, the analysis develops a comprehensive understanding of Olist's marketplace ecosystem.

Project Workflow

The project followed a structured data analytics workflow to transform raw data into meaningful business insights and actionable recommendations.

1. Data Understanding

Understanding the dataset structure, business context, and key analytical objectives.

2. Data Cleaning

Handling missing values, inconsistencies, duplicates, and incorrect data formats.

3. Data Preparation

Merging relevant datasets and creating derived metrics required for analysis.

4. Exploratory Data Analysis

Analyzing marketplace performance, customer behavior, delivery, products, sellers, geography, and payments.

5. Business Analysis

Identifying key patterns, performance gaps, opportunities, and factors associated with customer satisfaction.

6. Insight Generation

Converting analytical findings into evidence-based business insights.

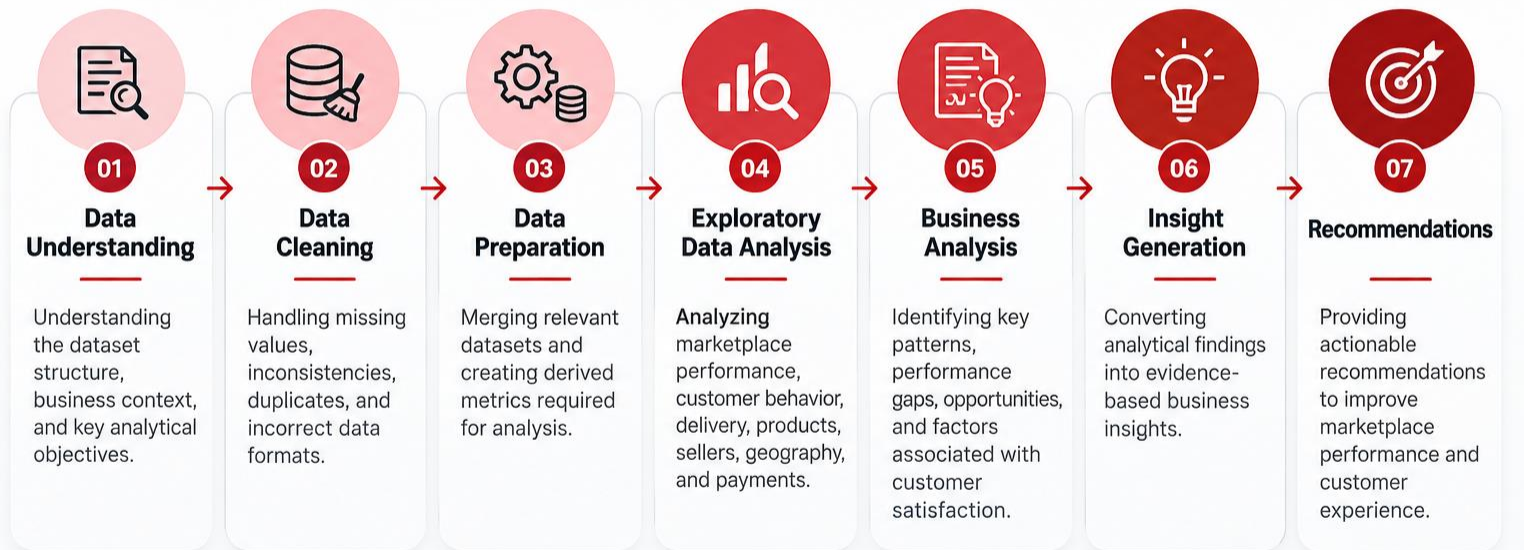
7. Recommendations

Providing actionable recommendations to improve marketplace performance and customer experience.

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PROJECT WORKFLOW

A structured approach to turn raw data into actionable business insights



From Data to Decisions → Better Customer Experience → Stronger Marketplace Performance

Data Understanding

Overview

The first step of the project was to understand the **business problem, dataset structure, and relationships between different datasets** before starting the analysis.

Since the Olist dataset consists of multiple interconnected tables, it was important to understand what information each dataset contains and how they connect with one another.

What was Done ?

The following activities were performed during the Data Understanding phase:

Understanding the Business Problem

The main objective was identified as understanding:

- Overall marketplace performance
- Customer satisfaction and dissatisfaction
- Delivery performance
- Seller performance
- Product category performance
- Geographic differences
- Payment behavior
- Customer retention

The analysis was designed to identify the factors most strongly associated with customer satisfaction and to provide actionable business recommendations.

Understanding the Dataset Structure

The dataset contains information across multiple business areas, including:

- Customers
- Orders
- Order Items
- Products
- Sellers
- Payments
- Customer Reviews
- Geographic Information

Understanding Dataset Relationships

The datasets were studied to understand how they connect through important identifiers such as:

- order_id
- customer_id
- customer_unique_id
- product_id
- seller_id

This helped create a clear understanding of how customer, order, seller, product, payment, delivery, and review information could be connected for analysis.

Identifying Key Variables

Important variables were identified for the analysis, including:

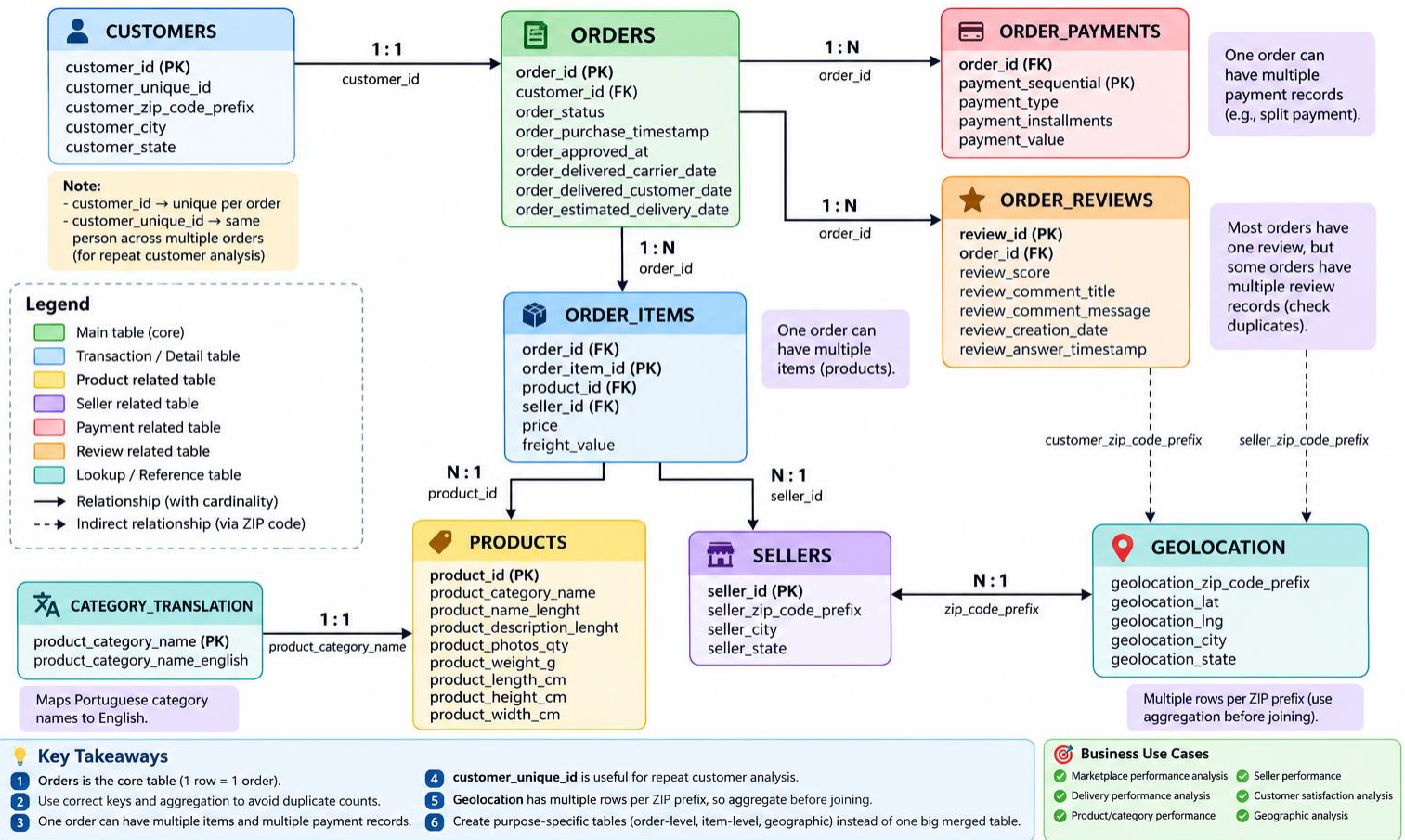
- Order status
- Purchase date
- Delivery date
- Estimated delivery date
- Review score
- Product category
- Payment type
- Payment value
- Seller location
- Customer state
- Product price
- Freight value



Olist E-commerce Dataset

Dataset Relationship Map

9 Datasets • 100K Orders • Sep 2016 – Oct 2018
Understand. Analyze. Find Insights. Drive Impact.



Data Cleaning

Overview

After understanding the dataset, the next step was to clean and standardize the data to ensure it was reliable and suitable for analysis.

Because the Olist dataset contains multiple tables and real-world transactional data, data quality checks were necessary before performing business analysis.

What Was Done

Handling Missing Values

Missing values were identified across relevant datasets and handled appropriately based on their importance to the analysis.

Checking Duplicate Records

The datasets were examined for duplicate records to avoid inaccurate calculations and double-counting.

Standardizing Data

Text fields such as city names and categories were cleaned and standardized to maintain consistency across datasets.

Correcting Data Types

Date and numerical columns were converted into appropriate formats to support calculations and time-based analysis.

Validating Data Quality

Important fields were checked for invalid, unknown, or inconsistent values. These records were either cleaned, standardized, or appropriately flagged.

Creating Clean Datasets

Separate cleaned versions of the datasets were created and used for further analysis. This ensured that the business insights were based on consistent and reliable data.

Data Preparation

Overview

After cleaning the individual datasets, the next step was to prepare the data for analysis by combining relevant tables and creating meaningful business metrics.

Since the Olist dataset is distributed across multiple interconnected tables, data preparation was essential to create a consolidated view of marketplace performance and customer experience.

What Was Done

Merging Relevant Datasets

Relevant datasets were connected using key identifiers such as:

- order_id
- customer_id
- product_id
- seller_id

This allowed different dimensions of the marketplace to be analyzed together.

Creating a Master Dataset

A consolidated analytical dataset was created by combining relevant information related to:

- Orders and order status
- Customers and geographic location
- Products and categories
- Sellers
- Payments
- Delivery information
- Customer review scores

Creating Derived Metrics

Additional business metrics were created to support the analysis, including:

- Delivery days
- Delivery status
- Late delivery rate
- Revenue
- Average order value
- Revenue percentage
- Customer segments
- Seller segments
- Performance categories

Preparing Analysis-Specific Datasets

Separate datasets and summaries were prepared for different areas of analysis, such as:

- Delivery performance
- Product category performance
- State-level performance
- Seller performance
- Payment behavior
- Customer retention and segmentation

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the overall patterns, trends, and relationships within the Olist marketplace data.

The analysis examined key business areas including:

- Order and revenue trends over time
- Customer review score distribution
- Delivery performance and delivery delays
- Product category performance
- Geographic differences across Brazilian states
- Seller performance and revenue contribution
- Payment method and installment behavior
- Customer purchasing and repeat purchase patterns

Visualizations and summary statistics were used to identify important trends, unusual patterns, and potential relationships between operational performance and customer satisfaction.

The EDA helped establish the foundation for deeper business analysis and root cause investigation in the following stages.

Marketplace Performance

Chart 1: Monthly Order Trend

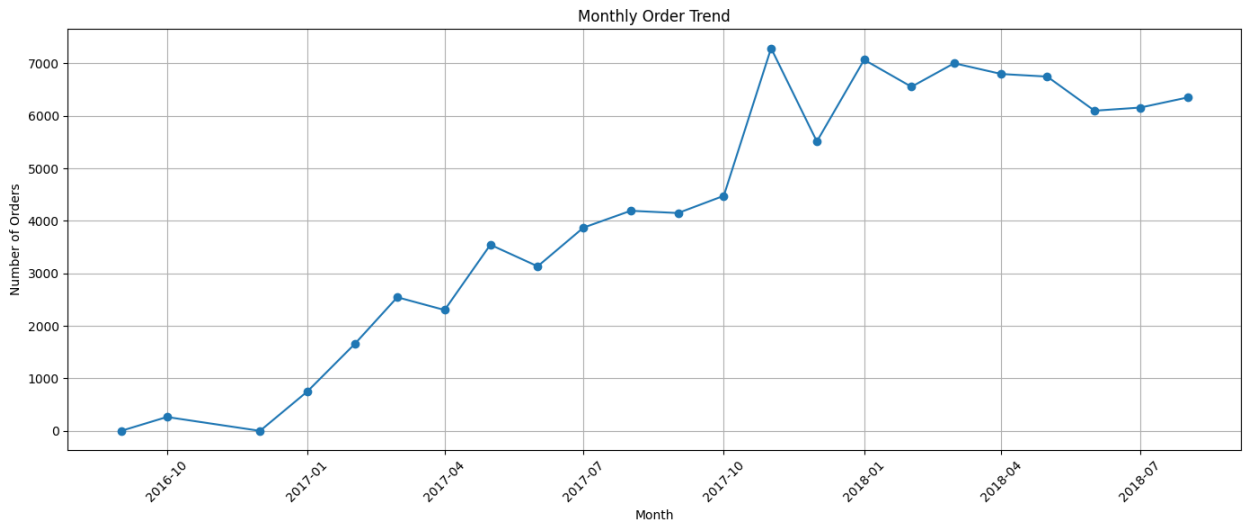
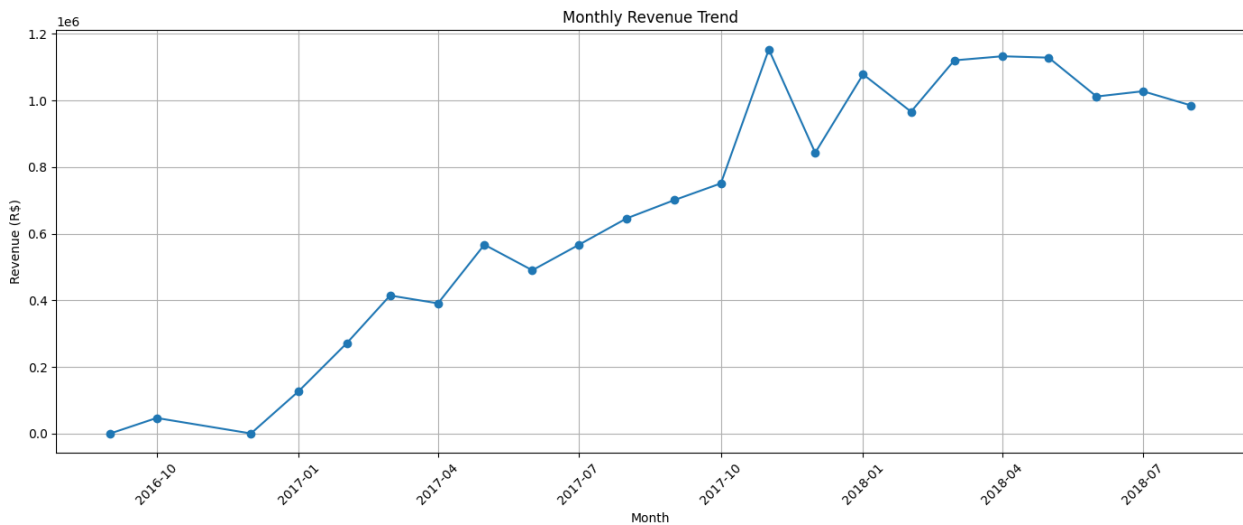


Chart 2: Monthly Revenue Trend



Key Finding

The Olist marketplace experienced strong growth in both order volume and revenue throughout 2017, reaching its highest levels in late 2017 and early 2018. After this rapid growth phase, marketplace performance remained relatively stable at a high level during 2018.

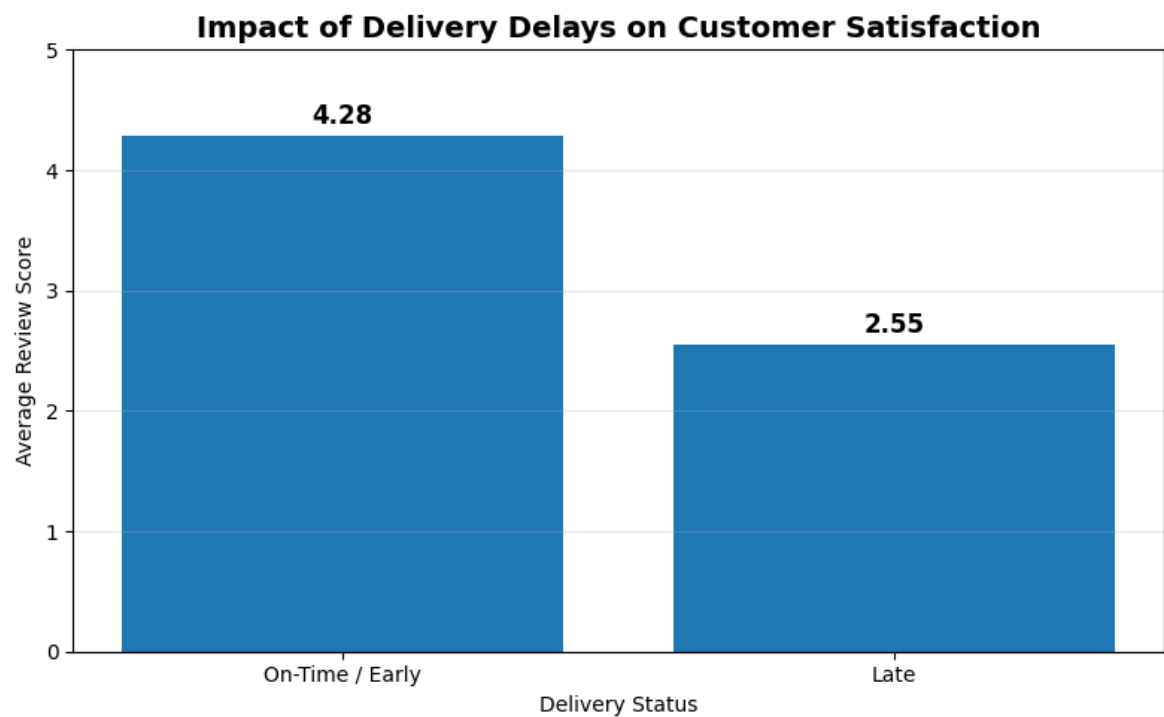
Average Order Value remained comparatively stable, generally between R\$145 and R\$175, indicating that overall revenue growth was primarily driven by an increase in the number of orders rather than a major increase in spending per order.

Business Impact

The marketplace successfully scaled its transaction volume and revenue while maintaining a relatively stable average order value. This means future growth opportunities depend strongly on increasing customer demand, improving operational capacity, and maintaining a positive customer experience as order volumes increase.

Delivery Performance vs Customer Satisfaction

Chart 1: Delivery Delay vs Customer Satisfaction



Key Finding

The analysis shows a strong relationship between delivery performance and customer satisfaction. Orders delivered on time or earlier received an average review score of **4.28**, while late orders received an average score of only **2.55**.

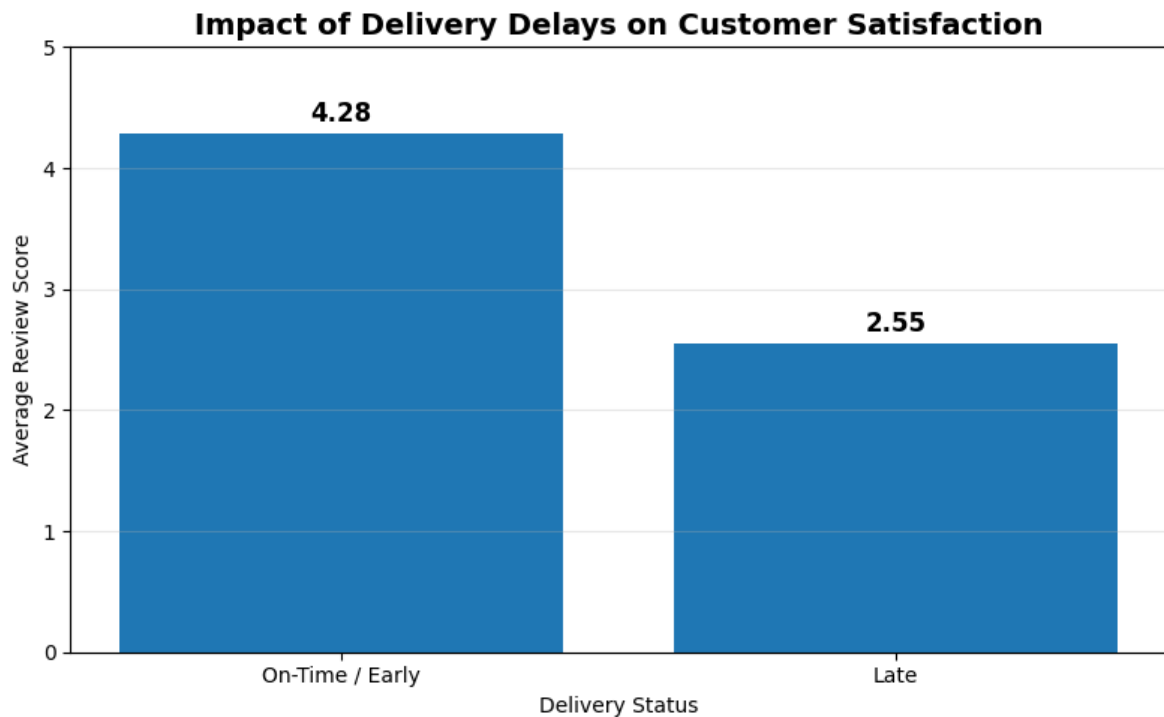
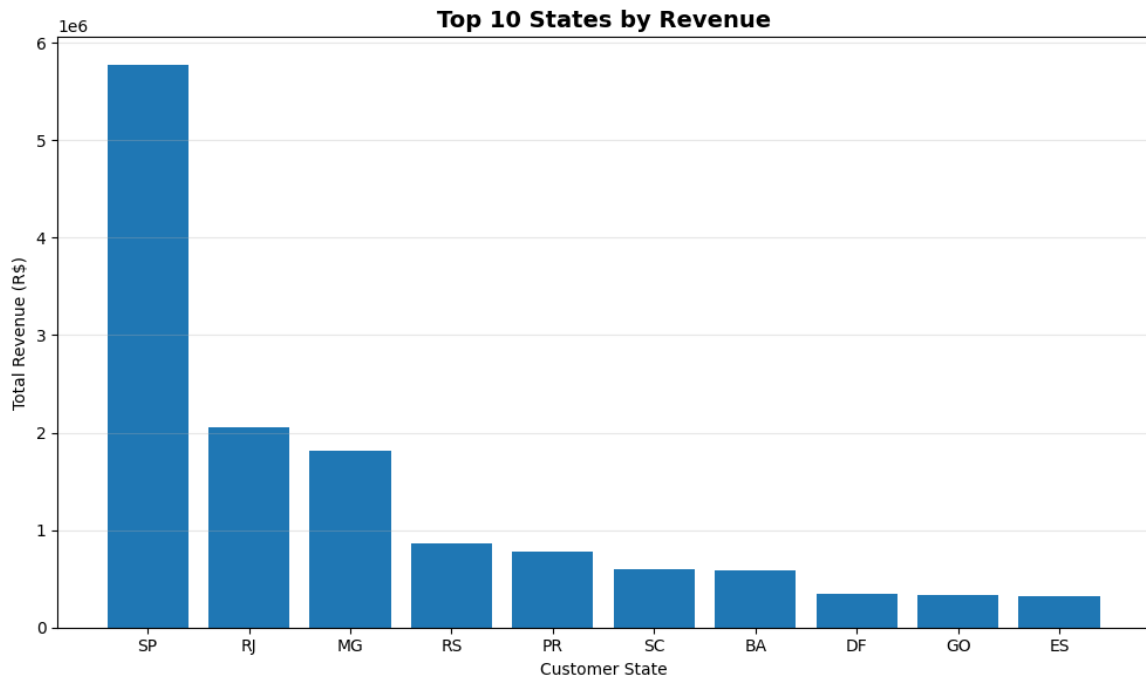
This represents a **1.73-point decline in customer satisfaction**, making delivery delays one of the strongest factors associated with negative customer reviews.

Business Impact

As Olist continues to scale, delivery reliability should be treated as a critical customer experience priority. Reducing late deliveries could have a significant positive impact on customer satisfaction, customer trust, and potentially long-term customer retention.

Geographic Performance Analysis

Chart 1: Revenue by Customer State



Key Finding

1. São Paulo (SP) Dominates the Marketplace

- 40,501 orders
- R\$5.77M revenue
- 37.41% of total revenue
- Average delivery: 8.76 days
- Average review score: 4.23

SP is the largest and strongest market in the dataset.

2. Regional Logistics Performance Varies Significantly

Several states show substantially longer delivery times and higher late-delivery rates.

Examples:

- **AL:** 24.54 average delivery days and 23.93% late delivery rate
- **MA:** 21.57 days and 19.67% late delivery rate
- **CE:** 21.27 days and 15.32% late delivery rate

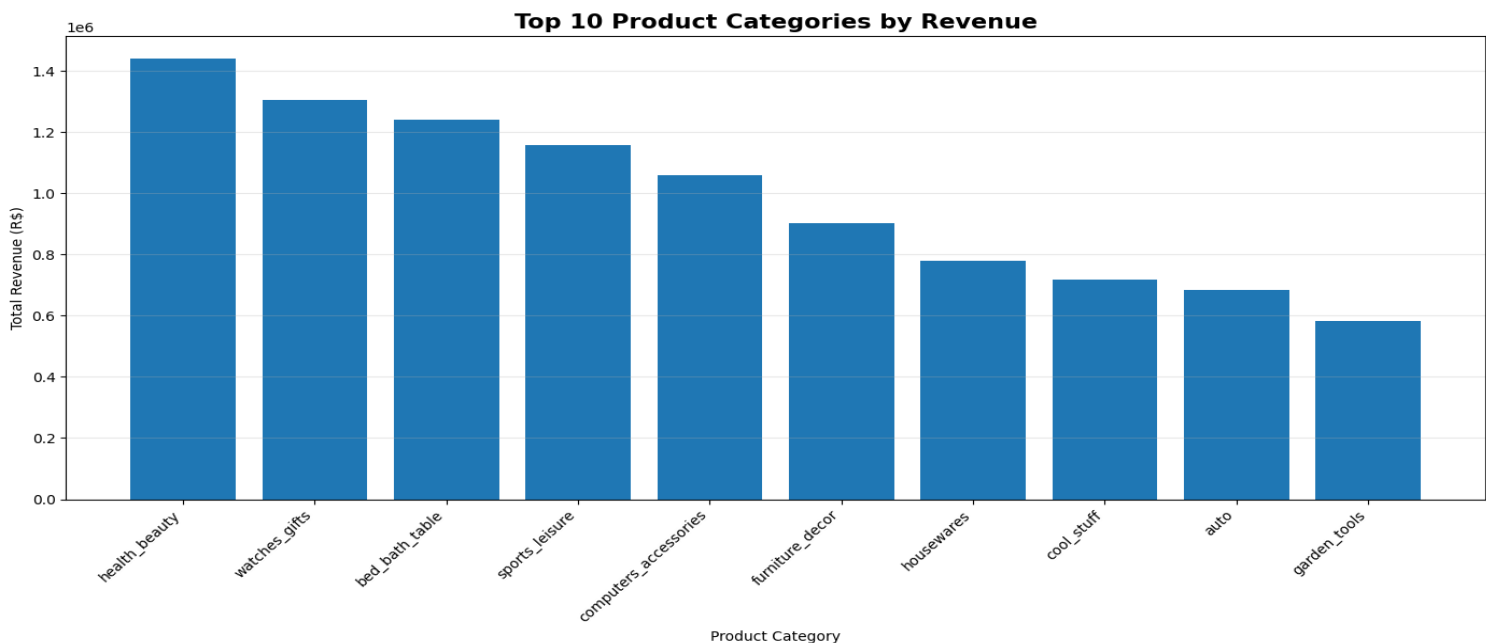
This indicates that customer experience is not consistent across regions.

Business Impact

Olist should prioritize logistics improvements in regions with consistently high delivery times and late-delivery rates. Improving regional fulfillment performance could reduce customer dissatisfaction and create a more consistent marketplace experience across Brazil.

Product Category Performance Analysis

Chart 1: Top Product Categories by Revenue



Key Finding

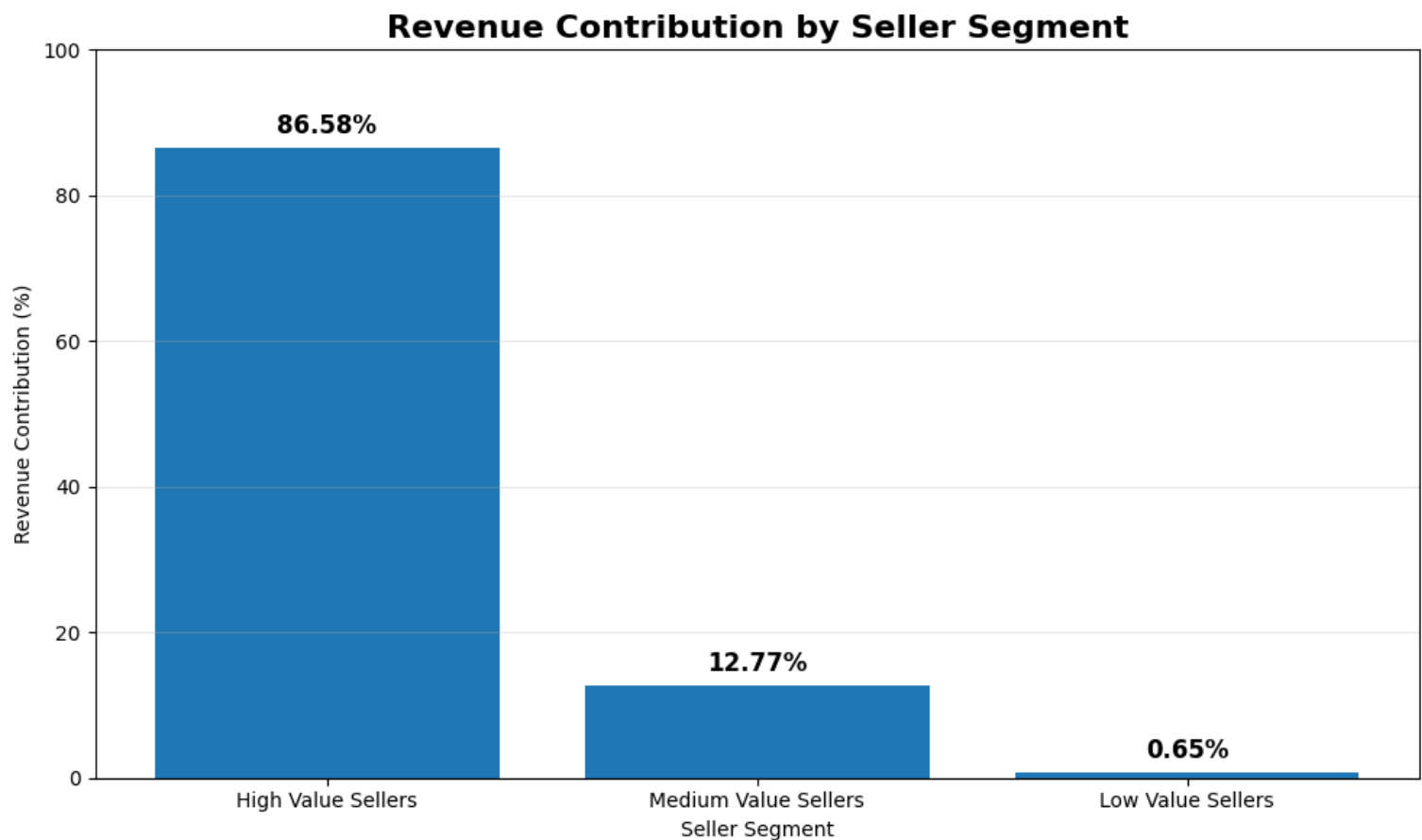
Product categories show significant differences in their contribution to marketplace revenue and customer experience. High-performing categories generate substantial revenue, while some categories may require attention due to weaker review scores or operational performance.

Business Impact

Olist should monitor categories using both financial performance and customer satisfaction, rather than focusing only on revenue. High-revenue categories with weaker customer satisfaction represent potential business risks because problems in these categories can affect a large number of customers.

Seller Performance Analysis

Chart 1: Revenue Contribution by Seller Segement



Key Finding

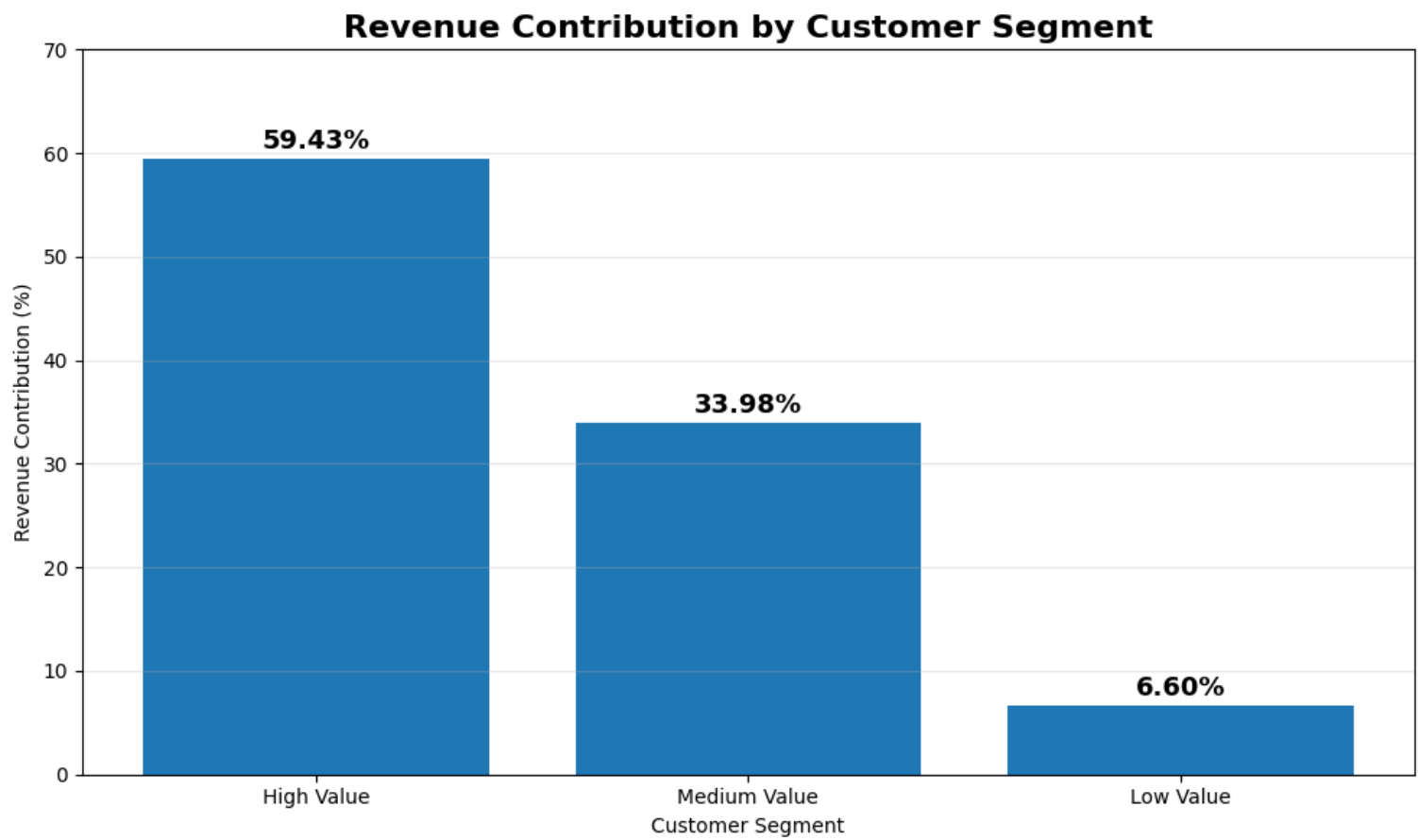
Seller revenue is highly concentrated among a relatively small group of high-value sellers. Although high-value sellers represent only approximately 25% of all sellers, they generate 86.58% of total seller revenue

Business Impact

Olist has a significant dependency on its high-value sellers. Maintaining strong relationships and monitoring the operational performance of these sellers is important because disruptions among this group could have a substantial impact on marketplace revenue.

Customer Analysis & Retention

Chart 1 Revenue Contribution by Customer Segment.



Key Finding

Customer revenue is concentrated among high-value customers, with **25% of customers generating approximately 59.43% of total revenue.**

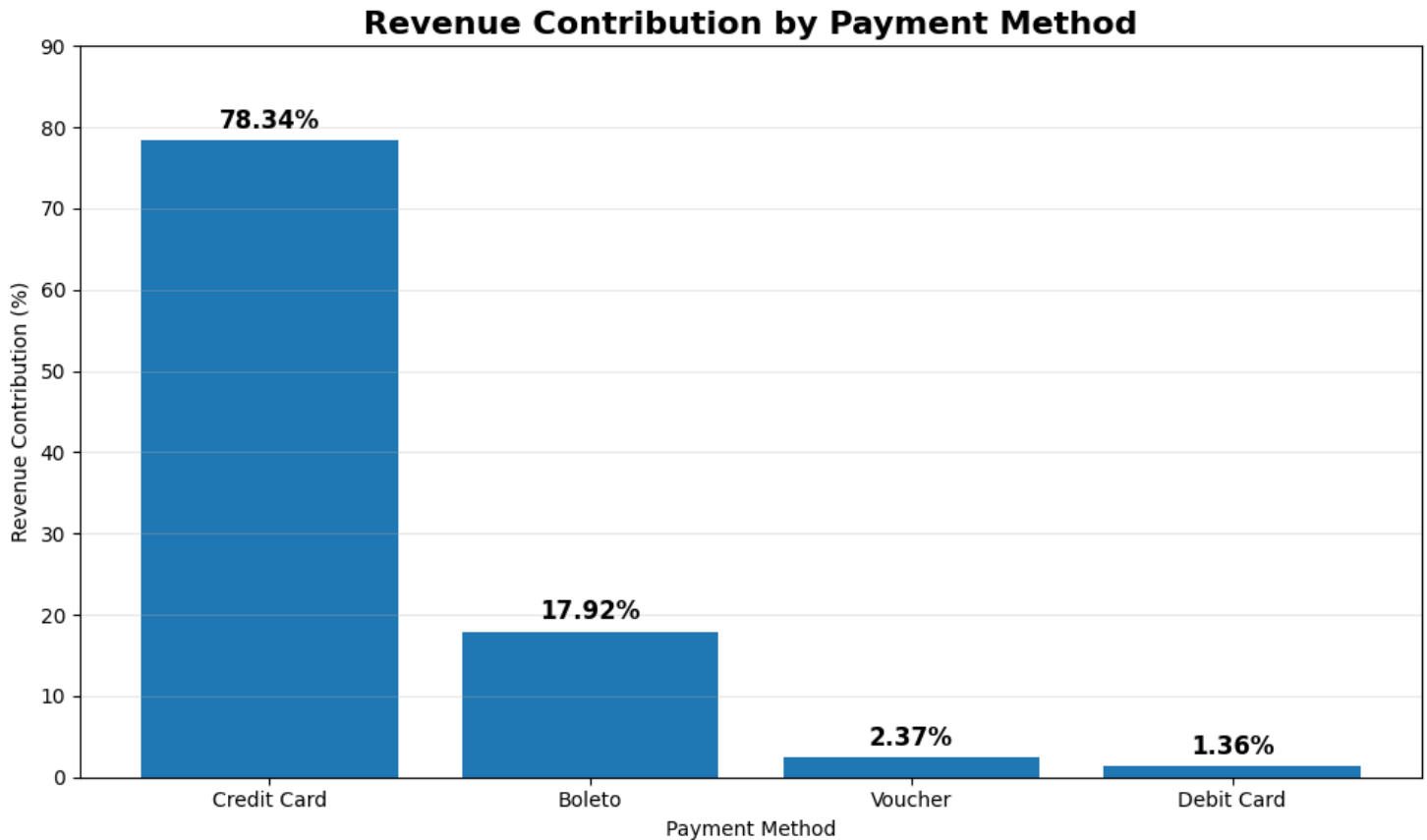
However, the repeat customer rate is only **3%**, indicating that most customers made only one purchase during the observed period.

Business Impact

The low repeat customer rate represents a major opportunity for Olist. Improving the customer experience, particularly delivery reliability, could help increase customer trust and encourage more customers to return to the marketplace.

Payment Behavior Analysis

Chart 1 Revenue Contribution by Payment Method



Key Finding

Credit cards are the dominant payment method, accounting for approximately **78.34% of total payment revenue**. Boleto is the second-largest payment method, contributing **17.92%**, while voucher and debit card payments represent relatively small shares.

Credit card payments also support installment purchases, with an average of approximately **3.5 installments per payment**.

Business Impact

Because credit cards dominate marketplace transactions, maintaining a smooth and reliable credit card payment experience is important for protecting the majority of marketplace revenue. Payment behavior appears to be more strongly related to transaction value and purchasing convenience than to customer satisfaction, making it a **secondary contributor** to the overall customer experience.

Insights Generations

Overview

After completing the business analysis, the findings were consolidated into clear and meaningful insights that directly address Olist's business challenges.

This phase focused on moving beyond individual charts and metrics to identify the **most important patterns, relationships, risks, and opportunities** within the marketplace.

What Was Done

Identifying Key Patterns

Results from different areas of analysis were compared to identify consistent patterns across:

- Delivery performance
- Customer satisfaction
- Product categories
- Geographic regions
- Sellers
- Payments
- Customer behavior

Connecting Multiple Findings

Instead of analyzing each metric separately, related findings were connected to develop a stronger business story.

For example, delivery performance was evaluated alongside customer review scores to understand its association with customer satisfaction.

Identifying Business Risks

Areas showing weaker performance were highlighted, including:

- High late-delivery rates
- Low customer satisfaction
- Underperforming categories
- Weak geographic markets
- Low customer retention
-

Identifying Opportunities

Strong-performing categories, markets, sellers, and customer segments were examined to identify potential growth opportunities and successful patterns that could be replicated.

Prioritizing Key Insights

The findings were ranked based on their potential business impact, helping distinguish between:

- Primary performance drivers
- Secondary contributing factors
- Business risks
- Growth opportunities

Actionable Recommendations

Overview

The final stage of the analysis focused on converting the key analytical findings into practical and actionable recommendations for Olist. The recommendations are prioritized based on their potential impact on customer satisfaction, customer retention, operational performance, revenue protection, and sustainable marketplace growth.

1. Improve Customer Retention

Finding

The analysis shows that approximately **97% of customers are one-time buyers**, while repeat customers represent a small proportion of the customer base.

Business Impact

The low repeat purchase rate highlights a significant opportunity to increase **Customer Lifetime Value (CLV)**. Improving retention could generate additional revenue without relying entirely on acquiring new customers.

Recommendations

- Launch targeted second-purchase campaigns.
- Introduce loyalty and retention programs.
- Provide personalized product recommendations.
- Create customer reactivation campaigns.
- Improve the post-purchase experience to encourage customers to return.

2. Prioritize Delivery Performance

Finding

Delivery performance shows the strongest observed relationship with customer satisfaction.

- **On-Time/Early Orders:** Average review score of 4.28
- **Late Orders:** Average review score of 2.55

This represents a **1.73-point drop in customer review scores**.

Business Impact

Late deliveries are strongly associated with lower customer satisfaction and may also negatively affect customer trust and future purchase behavior.

Recommendations

- Implement real-time delivery delay monitoring.
- Create early-warning systems for potentially delayed orders.
- Improve logistics performance in high-delay states.
- Monitor seller dispatch performance.
- Identify recurring logistics bottlenecks and address them proactively.

3. Protect High-Value Customers

Finding

Approximately **25% of High-Value customers generate 59.43% of total customer revenue.**

Business Impact

A relatively small group of customers contributes a significant share of marketplace revenue. Losing these customers could have a substantial financial impact.

Recommendations

- Develop a VIP or priority customer program.
- Provide personalized offers and recommendations.
- Offer exclusive benefits to valuable customers.
- Prioritize service recovery when High-Value customers experience operational problems.
- Build targeted retention strategies for high-value customer segments.

4. Address Geographic Performance Gaps

Finding

Marketplace performance varies significantly across Brazilian states.

São Paulo (SP) is the largest market, generating approximately **37.41% of total customer revenue.** Other regions experience longer delivery times and higher late-delivery rates.

Business Impact

Heavy concentration in one geographic market creates dependency risk, while inconsistent logistics performance across regions can lead to unequal customer experiences.

Recommendations

- Improve logistics infrastructure in high-delay regions.
- Expand strategically into high-potential states.
- Increase seller availability outside major markets.
- Develop region-specific operational strategies.
- Monitor geographic delivery performance continuously.

5. Strengthen Seller Performance Management

Finding

Seller revenue is highly concentrated. Approximately **25% of High-Value sellers generate 86.58% of total seller revenue.**

Business Impact

The marketplace is highly dependent on a relatively small group of sellers. Operational issues or seller churn within this group could significantly affect revenue.

Recommendations

- Retain and support High-Value sellers.
- Develop Medium-Value sellers to increase their contribution.
- Implement seller performance monitoring programs.

- Identify sellers requiring operational support.
- Reduce dependency on a small group of top-performing sellers.

6. Focus on High-Risk Categories and Operational Issues

Finding

Product categories vary in terms of revenue, sales volume, pricing, freight costs, and customer experience.

Business Impact

High-revenue categories with weaker operational performance or lower customer satisfaction could represent a significant revenue risk.

Recommendations

- Monitor high-revenue categories closely.
- Investigate categories with weaker customer satisfaction.
- Identify category-specific delivery or logistics problems.
- Protect strong-performing categories while improving weaker ones.
- Use category-level performance metrics for operational decision-making.

7. Maintain a Reliable Payment Experience

Finding

Credit cards are the dominant payment method, contributing approximately **78.34% of total payment revenue**.

Business Impact

A reliable credit card payment experience is important because a large majority of marketplace revenue depends on this payment method.

Recommendations

- Maintain payment reliability and transaction security.
- Monitor payment failures and transaction issues.
- Continue supporting installment-based purchasing.
- Optimize the checkout experience to reduce friction.

Project Outcome

This project delivers a comprehensive, evidence-based understanding of Olist's marketplace performance and customer experience.

The analysis identifies the key factors associated with customer satisfaction and dissatisfaction, with **delivery delays emerging as a major factor influencing review scores**. It also highlights important differences across **product categories, customer regions, sellers, payment methods, and customer segments**.

The project transforms raw e-commerce data into actionable business insights that can help Olist:

- Improve delivery and logistics performance.
- Reduce customer dissatisfaction caused by delayed orders.
- Identify high-performing and underperforming product categories.
- Address regional performance gaps.
- Monitor seller performance and marketplace concentration.
- Understand customer value and repeat purchase behavior.
- Prioritize business actions based on data-driven evidence.

Overall, the project provides a structured foundation for helping Olist improve customer experience, optimize marketplace operations, and support sustainable business growth.

Conclusion

The Olist Marketplace Performance and Customer Experience Analysis provides a comprehensive understanding of how different business and operational factors are associated with marketplace performance and customer satisfaction.

The analysis shows that **delivery performance is one of the strongest factors associated with customer satisfaction**. Customers receiving orders late had significantly lower review scores compared with customers whose orders arrived on time or early. This highlights logistics and delivery reliability as a critical area for improvement.

The study also identified meaningful variations across **product categories, geographic regions, sellers, payment methods, and customer segments**. While some categories and markets demonstrate strong performance, others show lower customer satisfaction, longer delivery times, and higher late-delivery rates.

Another important finding is that a relatively small group of high-value customers and sellers contributes a significant share of marketplace revenue. At the same time, the low customer repeat rate indicates a major opportunity for Olist to improve customer retention and encourage repeat purchases.

Overall, the analysis demonstrates that improving **delivery reliability, regional logistics performance, seller monitoring, and customer retention** should be key priorities for Olist. By using these insights to guide business decisions, Olist can improve the customer experience, strengthen marketplace performance, and support sustainable long-term growth.